
PaperMemory: Evidence-Packet Retrieval for Local-First Literature Assistance

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Abstract

PDF-based literature work can be slow because researchers need page-level evidence, citation support, and visible uncertainty before they can trust a draft answer. PaperMemory is a local-first compound AI assistant for postgraduate researchers who work with bounded PDF collections. It builds page-level EvidencePackets, combines visual and selectable-text retrieval, runs a bounded evidence-seeking loop, checks reliability, and keeps human verification in the workflow. Early market evidence from informal interviews with 10 PhD, master's, or postgraduate researchers gives a mean conditional purchase likelihood of 4.2/5 if the claimed functions are delivered, and 70% prefer PaperMemory-provided API/model access through a monthly subscription rather than BYOK-only setup. We therefore evaluate a freemium model with a free BYOK tier and proposed 10 and 20 USD/month tiers that include weekly capped, provider-compliant API relay for managed LLM access. On the local synthetic fixture suite, BM25 improves Recall@1 from 77.8% to 88.9% and retrieval-level refusal correctness from 33.3% to 100.0%; robustness fixtures cover missing evidence, citation drift, prompt-injection-like PDF text, conflicts, and low-text pages. A cost stress test estimates positive base-case net savings for first-pass evidence gathering under stated assumptions, while leaving conversion, retention, hosting cost, support cost, licensing, and real-corpus retrieval quality as open commercialization risks.

1 Introduction

Postgraduate researchers often spend early writing time turning PDFs into claims they can safely cite. The hard part goes beyond finding a relevant page. A useful research assistant must show why a page was used, which citation supports the sentence, and where the answer should stop because the evidence is thin. When that boundary is hidden, users still have to redo the evidence work by hand.

PaperMemory targets this paid workflow directly. In informal interviews with 10 university PhD, master's, or postgraduate researchers, the mean conditional purchase likelihood was 4.2/5 if PaperMemory delivered the promised functions: local PDF ingestion, evidence-backed answers, visible citations, bounded uncertainty, and a simpler model-access path. Seven of the 10 interviewees preferred PaperMemory-provided API/model access through a monthly subscription over a BYOK-only setup (PaperMemory, 2026a). Appendix B records the short questionnaire protocol and aggregate response framing. This is early willingness-to-pay evidence, not product-market fit, but it is enough to justify testing a paid model in this course-stage report.

Existing research assistants such as Elicit and Consensus are priced around evidence-oriented workflows, although they are market comparators rather than executable baselines for this report (Elicit, 2026; Consensus, 2026). PaperMemory's narrower position is local-first PDF work: the user brings

Table 1: A small postgraduate interview sample supports paid workflow interest, especially when PaperMemory provides managed model access, but the signal is early intent evidence.

Evidence item	Result	Boundary
Interview sample	10 postgraduate interviewees	Small interview sample, not a representative survey.
Conditional purchase likelihood	4.2 / 5 mean	Applies only if the claimed PDF, evidence, citation, and uncertainty workflow is delivered.
API/model access preference	70%	Preference for product-provided API/model access through monthly subscription over BYOK-only setup.
Proposed packaging	Free BYOK plus 10 and 20 USD/month tiers	Proposed revenue design, not actual conversion, retention, or production revenue.

Table 2: Major claims, evidence, and boundaries visible to graders inside the PDF.

Claim	Evidence in this report	Boundary
Market signal for a paid postgraduate workflow.	Interview aggregate: n=10 postgraduate researchers, 4.2/5 conditional purchase likelihood, and 70% managed-access preference.	Convenience sample; not product-market fit, paid conversion, retention, or validated revenue.
PaperMemory is a compound AI system rather than a chat-only UI.	EvidencePackets, visual and text retrieval paths, BM25, hybrid page fusion, bounded evidence loop, reliability status, and evidence-visible UI.	Bounded local agent; not general AGI or full literature-review automation.
Retrieval components work on the local fixture suite.	Keyword, visual API, BM25, and hybrid rows; BM25 reaches 88.9% R@1 and 100.0% retrieval-level refusal correctness on the synthetic fixture.	Synthetic/local fixture; not real-corpus superiority.
Subscription packaging has bounded cost logic.	Node 9 10-PDF/30-PDF cost stress test plus provider-compliant API relay sensitivity.	Planning estimate; excludes hosting, support, licensing, conversion, retention, and usage telemetry.
Trust behavior is implemented as visible limits.	Robustness matrix, failure gallery, reliability labels, citation filtering, and prompt-injection-like PDF text boundary.	Deterministic regression evidence; not comprehensive security or OCR robustness.

37 a scoped collection, the system builds page-level EvidencePackets, and the interface exposes cita-
38 tions, limits, and reliability status before the user turns the draft into research writing.

39 The contribution is both technical and commercial. Technically, PaperMemory combines page ren-
40 dering, selectable-text manifests, BM25 retrieval, page-canonical rank fusion, EvidencePacket vali-
41 dation, bounded evidence seeking, reliability checks, and an evidence-visible UI. Commercially, it
42 connects that workflow to a freemium BYOK product and proposed paid tiers that remove API setup
43 friction through compliant managed model access. The evaluation remains bounded: synthetic/local
44 metrics and controlled live tests show implementation behavior, not broad scholarly-corpus superi-
45 ority.

46 2 Product Boundary and Revenue Logic

47 The first target users are postgraduate researchers who work with dense PDF collections and need
48 a faster first pass over evidence. A second group is small research, consulting, or R&D teams that
49 need traceable evidence packets before writing memos or reports. PaperMemory is not a systematic-
50 review replacement or final expert adjudicator. It supports evidence gathering, citation packaging,
51 and uncertainty surfacing before human review.

52 The proposed packaging follows the interview signal. The free tier is BYOK with limited groups,
53 which lowers adoption friction and keeps model cost outside the product. The 10 USD/month tier
54 adds provider-compliant API relay for managed LLM access with a lower weekly managed-model

Table 3: The proposed freemium model connects interview preference for managed model access with bounded usage costs and evidence-workflow value.

Tier	User value	Model/API handling	Cost boundary
Free BYOK	Limited groups for local PDF evidence work and adoption testing.	User supplies their own key or local model path.	No included model spend; product cost is mainly local workflow support.
10 USD/month	Lower-friction postgraduate workflow with more groups and routine evidence runs.	Provider-compliant API relay for managed LLM access under a lower weekly cap, BYOK fallback, and rate limits.	Plausible only if managed usage stays bounded and high-use cases are throttled or upgraded.
20 USD/month	Larger workspace for heavier literature projects and repeated report drafting.	Provider-compliant API relay under a higher weekly cap, larger workspace, and the same evidence controls.	Still requires fair-use policy, production hosting data, support-cost tracking, and licensing review.

boundary for routine postgraduate work. The 20 USD/month tier offers a larger workspace and a higher weekly managed-model boundary for heavier literature projects. The paid design assumes commercial API access or equivalent provider permission; it does not rely on consumer-account pooling, coding-plan resale, or unlimited inference. The exact quotas should be set only after usage telemetry exists; this report intentionally does not invent them.

The value logic is that users pay for convenience and evidence discipline together. The subscription removes API-key setup for the 70% of interviewed users who preferred managed access, while the product keeps the local-first evidence workflow visible: pages, citations, limits, and reliability status remain part of the answer. That distinction matters because a plain hosted chatbot can be cheaper to build, but it does not give the user the same packet-level audit trail.

The cost model supports a plausible subscription experiment under bounded usage. The Node 9 stress test estimates API/compute costs of \$0.19 for a 10-PDF evidence packet, \$0.68 for a 30-PDF project evidence scan, and \$1.62 for systematic-review pre-screening, with positive base-case net savings in all three scenarios after operation, verification, API/compute, and scenario license assumptions (PaperMemory, 2026n). These numbers do not prove margins. They exclude production hosting, support, abuse controls, full licensing decisions, sales cycle cost, usage variance, actual conversion, and retention. The 10 and 20 USD/month tiers are therefore commercially plausible only under fair-use limits, and they remain proposed packaging rather than validated revenue.

3 Compound AI System Architecture

PaperMemory is a compound AI system over a local PDF collection. The ingestion path renders PDFs into page artifacts and extracts selectable text where the PDF permits it. Each page can therefore carry an image, a caption or text snippet, a selectable-text manifest entry, and a quality label such as low-text or OCR-needed. This split matters because visual retrieval can still use page layout, while BM25 can use exact terms when text exists.

Retrieval has three implemented paths. The visual path retrieves page candidates from the page-image evidence surface. The BM25 path indexes the text manifests with $k_1 = 1.5$ and $b = 0.75$. The hybrid path applies page-canonical reciprocal rank fusion with $k = 60$, then collapses visual and text hits onto the same $(paper, page)$ unit. That canonical page key keeps the downstream citation grain stable even when the two retrievers disagree about score scale or rank source.

The EvidencePacket schema is the system boundary between retrieval and generation. A packet records the query, paper scope, accepted EvidenceUnits, accepted page citations, rank traces, source modality, validation state, and limits. The public packet excludes local filesystem paths and keeps stable evidence ids so that chat, trace logs, and report material can point back to the same page units (PaperMemory, 2026f,j). Generation is allowed to see packet context, but final citations are filtered back to accepted packet pages.

Local PDFs	Retrieval and packet	Reliability layer	Verified outputs
User-selected paper scope	Page images, captions, selectable-text manifests, visual search, BM25, and page-canonical RRF into EvidencePackets	Requirement planning, coverage check, claim verification, and reliability status	Scoped chat, UI evidence panel, public trace, report-ready citations, and cost-model inputs
PDFs → page images/text manifest → visual/BM25/hybrid retrieval → EvidencePacket → bounded orchestrator → reliability report → chat/report artifacts			

Figure 1: PaperMemory pipeline with the reliability layer. The figure describes the report package over existing local artifacts; it does not add OCR, dense text retrieval, billing, or unrestricted agent actions.

Table 4: Bounded agent trace states. The agent can retry for missing evidence, but server checks keep actions inside the selected paper scope.

State	Agent action	Tool or component	Stop or transition	Safety boundary
Rewrite	Prepare a scoped retrieval query from the user turn.	Server planner hint.	Move to first retrieval.	Planner output is validated and budget-clamped.
Retrieve	Retrieve page evidence for the selected papers.	Hybrid, visual, or BM25 retrieval.	Build accepted EvidencePacket units.	Retrieval stays inside active paper scope.
Analyze	Compare packet evidence with required claims.	Requirement planner and coverage check.	Stop as <code>sufficient</code> or request targeted retry.	Missing evidence is recorded as a limit.
Retry	Search for uncovered requirements.	Coverage-gated retrieval retry.	Add only new accepted page units.	Pass count and query count are fixed by server limits.
Check	Check whether retry changed the packet.	Evidence delta and dedupe check.	Stop as <code>no_new_evidence</code> when delta is zero.	Repeated retrieval cannot loop forever.
Answer	Generate with final packet context and cite accepted pages.	Chat service and claim verifier.	Return answer plus reliability status.	Citations outside packet pages are stripped.

90 The reliability layer sits after packet construction and before the answer is treated as ready for the
91 user. It plans evidence requirements from the question, checks whether the selected papers and claim
92 types are covered, verifies answer claims against accepted evidence, and returns a visible `strong`,
93 `partial`, or `insufficient` status. The status is a review signal, not a correctness certificate. It
94 tells the user whether the packet supports the answer, where coverage is thin, and when the system
95 should withhold or narrow a claim (PaperMemory, 2026b).

96 This architecture also connects to the product cost model. Reliability checks add planner and veri-
97 fication calls when enabled, while disabled reliability mode avoids that extra work. The cost stress
98 test therefore belongs in the architecture story as a deployment control: stronger grounding can be
99 purchased with more model work when the task warrants it, and lighter runs remain available for
100 lower-risk exploration (PaperMemory, 2026n).

101 4 Agentic Autonomy and Implementation Logic

102 PaperMemory’s autonomy is bounded and server-verified. The agent does not browse freely, execute
103 arbitrary tools, or attempt a complete literature review. It follows a small state machine over the
104 selected paper scope. Planner output is treated as an untrusted hint, then the server clamps pass
105 budgets, query counts, and final evidence size before retrieval or answer generation can proceed.

106 The implemented loop starts with query rewrite. The system prepares a conversation-aware retrieval
107 query, then runs first retrieval through the selected retrieval mode, normally hybrid for scoped chat.

Table 5: Synthetic retrieval metrics from existing local artifacts. Higher is better for measured percentage columns.

Run	R@1	R@3	R@5	Citation	Refusal	Boundary
Keyword overlap baseline	88.9%	100.0%	100.0%	100.0%	33.3%	Same synthetic fixture; token overlap only, not LLM, BM25, VisRAG, or real corpus.
Visual API baseline	77.8%	100.0%	100.0%	100.0%	33.3%	Synthetic smoke corpus; not a real PDF or real VisRAG result.
BM25 text manifest	88.9%	100.0%	100.0%	100.0%	100.0%	Synthetic BM25-only text manifest fixture.
Hybrid page fusion	77.8%	100.0%	100.0%	100.0%	33.3%	Synthetic page-canonical fusion; 27 units carry both traces.

108 Evidence analysis inspects the accepted packet units and records missing requirements. If coverage
 109 is incomplete and the budget remains, the orchestrator issues a targeted second retrieval rather than
 110 answering immediately. The retry is coverage-gated: it must seek missing evidence, deduplicate
 111 accepted units, and report whether the new pass changed the packet.

112 Stop reasons are explicit. A `sufficient` trace can answer after the first pass when the packet covers
 113 the question. A `no_new_evidence` trace stops after a retry produces no new accepted units. The
 114 answer step then uses the final packet, strips citations that do not match accepted pages, and attaches
 115 the reliability status from the requirement, coverage, and claim-verification checks. A `partial`
 116 answer can still be useful when it says which part is supported. An `insufficient` answer asks the
 117 user to revise scope or add evidence instead of filling the gap with unsupported prose.

118 The live controlled run exercises the same boundary. The successful image-context chat re-
 119 turned HTTP 200, six evidence rows, six citations, three included evidence images, and a final
 120 `sufficient` stop reason. The missing-evidence chat used a bounded second retrieval and ended
 121 with `no_new_evidence`, while the answer stated that the evidence packet did not support the re-
 122 quested mitochondrial ribosome profiling claim. The prompt-injection-style page stayed evidence
 123 rather than instruction: the answer treated the page text as untrusted and did not reveal keys, execute
 124 tools, or alter citations (PaperMemory, 2026k,c).

125 5 Baselines, Evaluation, and Technical Results

126 The evaluation is a proof-of-capability package built from synthetic fixtures, deterministic regression
 127 tests, and a controlled local browser/API run. It is not a broad benchmark over scholarly PDFs. The
 128 synthetic retrieval suite has four papers, 12 pages, 12 golden questions, and three negative cases.
 129 The controlled local run used uploaded fixture PDFs and records UI flow, retrieval quality checks,
 130 bounded agent traces, and trust behavior. These artifacts support implementation claims, not claims
 131 about general corpus performance.

132 The baseline comparison has four measured synthetic retrieval rows. The first row is a deliberately
 133 weak keyword-overlap baseline on the same 12 golden questions. It scores title and caption to-
 134 kens, giving Recall@1 of 88.9%, Recall@3 and Recall@5 of 100.0%, citation-page correctness of
 135 100.0%, and retrieval-level refusal correctness of 33.3%; it is executable but has no EvidencePacket,
 136 model reasoning, BM25 weighting, visual retrieval, or sufficiency gate (PaperMemory, 2026o). The
 137 visual API baseline has Recall@1 of 77.8%, Recall@3 and Recall@5 of 100.0%, citation-page cor-
 138 rectness of 100.0%, and retrieval-level refusal correctness of 33.3%. The BM25 text-manifest row
 139 improves refusal correctness to 100.0% on the same synthetic setup, while preserving 100.0% Re-
 140 call@3, Recall@5, and citation-page correctness (PaperMemory, 2026e,h). GPT-only chat, Elicit,
 141 and Consensus remain qualitative market or workflow comparators rather than measured same-
 142 fixture retrieval baselines.

143 The hybrid page-fusion row should be read as provenance evidence. Its aggregate synthetic retrieval
 144 percentages match the visual baseline, but 27 returned units carry both visual and BM25 traces.

Table 6: Deterministic robustness package. Each case is a local regression fixture, not an adversarial benchmark.

Case	Expected behavior	Boundary
Empty scope	Conversation-mode limit and paper-style citation stripping.	No proof that all unsupported prose is detected.
Missing evidence	Partial packet with no accepted citations.	Covers empty accepted packets for known scope.
No-new-evidence loop	Stop reason records zero evidence delta.	Page dedupe, not semantic novelty.
Prompt-injection-like PDF text	PDF text remains untrusted evidence; unsafe planner hints are redacted.	Pattern-based deterministic phrases.
Citation drift	Unsupported page labels are stripped.	Simple <code>paper_id</code> p.N labels.
Conflicting numeric evidence	Verification-needed limit is surfaced.	Numeric heuristic only.
Low-text marker	Visual-first limit warns against invented missing text.	OCR remains future work.

145 That result confirms that page-canonical RRF can merge retriever evidence into a single packet unit
 146 without inventing a new citation grain (PaperMemory, 2026i). It does not show that hybrid retrieval
 147 is better on real paper collections.

148 The bounded evidence agent is evaluated through traces and regression behavior rather than Re-
 149 call@k. The trace examples show accepted evidence ids, pass indexes, evidence deltas, stop reasons,
 150 and final packet limits. The controlled live run adds a more realistic smoke test: search and chat
 151 reached the local API, evidence cards appeared in the UI, citation chips were rendered, missing ev-
 152 idence stopped safely, and prompt-injection-style PDF text remained inside the evidence boundary.
 153 The main technical result is therefore narrower than a benchmark claim: the current implementation
 154 connects retrieval, EvidencePackets, bounded retries, reliability status, and UI evidence surfacing in
 155 a reproducible local workflow.

156 6 Trust, Safety, and Robustness

157 Trust in PaperMemory depends on visible failure behavior. A research assistant can be useful only if
 158 the user can see when the evidence packet is strong, partial, or insufficient. PaperMemory therefore
 159 treats every answer as a product of selected papers, accepted EvidenceUnits, packet limits, and a
 160 reliability status. The status is a review signal for the user. It is not a correctness certificate.

161 The robustness package covers common ways a PDF-grounded assistant can fail. With an empty
 162 scope, PaperMemory enters conversation mode and strips paper-style citations because no selected
 163 paper evidence exists. With missing scoped evidence, the packet is partial, carries no accepted
 164 citations, and removes generated page labels that are not supported by the accepted packet. With
 165 repeated retrieval, a second pass that finds the same page stops with `no_new_evidence` rather than
 166 treating a duplicate as new support.

167 Prompt-injection-like PDF text is handled as untrusted evidence. The page text may appear inside
 168 the accepted evidence section, but planner prompts label it as evidence rather than instruction, ex-
 169 tra action fields are rejected, and unsafe planner hints are removed from public traces. Citation
 170 drift is handled at serialization time: citations outside accepted packet pages, including unknown-
 171 paper labels, are stripped before the answer reaches the user. Conflicting numeric evidence adds a
 172 verification-needed limit instead of forcing a single value. Low-text, empty, or OCR-needed pages
 173 add a visual-first limit so the assistant does not invent text that was not extracted (PaperMemory,
 174 2026l,m).

175 The reliability layer turns these cases into user-facing states. A `strong` answer has the planned
 176 evidence requirements covered by accepted pages. A `partial` answer can be useful, but the limits
 177 explain which evidence is missing or contested. An `insufficient` answer tells the user to change
 178 scope or add sources before treating the answer as paper-grounded. This is a practical review work-

Table 7: Node 9 planning estimates under stated assumptions. API/compute cost is per scenario, and savings are net USD after operation, human verification, API/compute, and license assumptions.

Scenario	PDFs	API/compute	Low	Base	High
10-PDF evidence packet	10	0.19	52.35	183.70	334.76
30-PDF project evidence scan	30	0.68	79.32	561.61	1175.45
Systematic-review pre-screening	200	1.62	-232.20	3423.86	7863.37

Table 8: Proposed pricing tiers for a subscription experiment. The tiers avoid exact quotas until usage telemetry exists.

Tier	Access model	Usage boundary	Human role
Free BYOK	User provides an API key or local model path for limited local PDF groups; no included model spend.	BYOK usage remains outside PaperMemory-managed LLM spend, with local limits on groups and requests.	User verifies pages and manages their own model setup.
10 USD/month	Provider-compliant PaperMemory API relay for users who prefer subscription access over BYOK-only setup.	Lower weekly managed-model cap, BYOK fallback above cap, rate limits, and fair-use controls.	User verifies cited pages before using claims in writing.
20 USD/month	Provider-compliant API relay with a higher weekly managed-model boundary.	Higher weekly cap than the 10 USD tier, still with BYOK fallback, rate limits, and fair-use controls.	User remains responsible for expert judgment and final citation checks.

flow for controlled local fixtures and live smoke tests. It is not a broad security result, an OCR result, or proof that every unsupported claim will be detected (PaperMemory, 2026b).

7 Commercial Stress Test

The commercial question is whether evidence-packet work can save enough expert time to justify a paid workflow while keeping human verification in place. Node 9 models that question with:

$$\text{net savings} = (h_m - h_p - h_v)r - c_a - c_l,$$

where h_m is manual work, h_p is PaperMemory operation time, h_v is human verification time, r is a fully loaded hourly rate, c_a is API or compute cost, and c_l is a license allocation. The formula makes verification a cost, not an optional afterthought.

The per-scenario API or compute cost is small in the current stress test: 0.19 USD for a 10-PDF evidence packet, 0.68 USD for a 30-PDF project evidence scan, and 1.62 USD for systematic-review pre-screening. Base-case net savings are 183.70 USD, 561.61 USD, and 3423.86 USD after operation, verification, API/compute, and license assumptions. The low case for systematic-review pre-screening is -232.20 USD, which matters. It shows that the value case can fail when verification time and deployment assumptions dominate.

The interview signal supports a subscription experiment rather than a BYOK-only product. Seven of 10 interviewed postgraduate researchers preferred PaperMemory-provided API/model access through a monthly subscription. Table 9 uses the 10-PDF and 30-PDF Node 9 API/compute estimates as illustrative managed-access stress cases, not production quotas or validated margins. The proposed packaging keeps the free tier as BYOK, then tests 10 USD/month and 20 USD/month managed-access tiers through compliant API relay. Those tiers are plausible only with weekly capped managed LLM usage, BYOK fallback, rate limits, or a fair-use policy. Exact quotas should come from usage telemetry, not this report.

The paid tiers bundle compliant managed LLM access support rather than unlimited inference. The weekly cap is the key margin control: once a user exceeds the managed allowance, the product

Table 9: Illustrative subscription stress test for provider-compliant managed-LLM API relay. Costs reuse the routine 10-PDF and 30-PDF Node 9 estimates and are planning assumptions, not production quotas or validated margins.

Tier	Modeled month	API cost and headroom	Boundary
Free BYOK	User-supplied model/key.	0 USD PaperMemory model spend; 0 USD subscription revenue.	Adoption tier; group/request limits control local load, and human verification remains required.
10 USD/month	Four 10-PDF evidence-packet runs.	$4 \times 0.19 = 0.76$ USD; 9.24 USD before non-model costs.	Light-use stress case, not a promised quota; over-cap use falls back to BYOK or wait.
20 USD/month	Four 30-PDF evidence-scan runs.	$4 \times 0.68 = 2.72$ USD; 17.28 USD before non-model costs.	Heavier-use stress case with weekly caps, rate limits, and fair-use review.
Heavy-use guard	Eight 30-PDF evidence-scan runs.	$8 \times 0.68 = 5.44$ USD; 14.56 USD before non-model costs.	Shows why unlimited inference or account-pool resale is not offered.

can pause, ask the user to wait, or fall back to BYOK. This keeps the commercial claim tied to a bounded service design rather than an unlimited API subsidy or subscription-account arbitrage. The stress test does not prove paid conversion, retention, production margins, hosting cost, support cost, abuse cost, or legal cost. It also does not make PaperMemory a systematic-review replacement. The stronger commercial claim is narrower: under the current assumptions, API cost is not the main blocker for first-pass evidence gathering. The product risk sits in user adoption, verification effort, usage variance, and deployment licensing.

8 Limitations, Ethics, and Reproducibility

The evaluation package is local and controlled. Retrieval metrics come from synthetic fixtures, not a broad scholarly PDF benchmark. The live test uses a small controlled corpus and includes one duplicate-upload pollution caveat. Text extraction is selectable-text only; low-text and scanned pages are flagged, but OCR is future work. The text path uses BM25 and page-canonical fusion, not dense semantic text retrieval. The reliability layer checks packet support for answer claims, but it does not establish exhaustive literature coverage.

The market evidence is also early. The interview sample has 10 postgraduate researchers, a mean conditional purchase likelihood of 4.2/5, and a 70% preference for managed API/model access. The appendix records the questionnaire protocol, but the report uses only aggregate notes. Those numbers justify a product experiment, not a claim about paid adoption. The planned 10 USD/month and 20 USD/month tiers still need pricing tests, usage telemetry, support estimates, and churn data.

Ethically, PaperMemory should remain a first-pass assistant. It can reduce citation packaging work, but users still need to inspect cited pages before academic, medical, legal, or policy use. The local-first and BYOK options reduce some centralization risk, yet external model APIs may still receive prompts and evidence snippets if the user configures them. Public traces therefore avoid local paths and unsafe planner hints, and generated citations are constrained to accepted packet pages.

Reproducibility materials live under the active repository root `D:/codex/papermemory`. The older `D:/codex/llm_paper_assis/papermemory` path is now a junction to the same working tree. Appendix A records the branch, baseline environment, LaTeX command, and verification snapshots. Appendix B records the interview sample, questionnaire protocol, aggregate results, and limits. Appendix D records the robustness gallery and cost assumptions. A separate three-minute demo video accompanies the submission; the report focuses on technical evidence, market proof, cost stress testing, and reproducibility.

9 Conclusion

PaperMemory addresses a concrete research workflow: turning a local PDF collection into cited first-pass evidence that a human can inspect. The current system is more than a chat box because

237 it combines visual and text retrieval, page-canonical fusion, EvidencePackets, bounded evidence
238 seeking, reliability status, and UI evidence surfacing. The strongest evidence is still local: synthetic
239 retrieval fixtures, deterministic robustness tests, a controlled live run, early postgraduate interviews,
240 and a cost stress test.

241 The next steps are practical rather than rhetorical. PaperMemory needs real local-corpus evaluation,
242 OCR and dense text retrieval experiments, usage telemetry for the proposed subscription tiers, and
243 clearer licensing decisions before deployment. Until then, the right claim is bounded: PaperMem-
244 ory is a plausible local-first assistant for first-pass evidence gathering and citation packaging, with
245 visible limits and human verification built into the workflow.

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279 [reproducibility_environment.md](reports/final/results/reproducibility_environment.md).
- 280 PaperMemory local artifact. Baseline retrieval metrics. [reports/final/results/baseline_](reports/final/results/baseline_metrics.md)
281 [metrics.md](reports/final/results/baseline_metrics.md).

282 PaperMemory local artifact. Keyword overlap baseline metrics. `reports/final/results/`
283 `keyword_baseline_metrics.md`.

284 PaperMemory local artifact. Evidence contract. `reports/final/results/evidence_`
285 `contract.md`.

286 PaperMemory local artifact. Text manifest quality. `reports/final/results/text_manifest_`
287 `quality.md`.

288 PaperMemory local artifact. BM25 text retrieval metrics. `reports/final/results/bm25_`
289 `metrics.md`.

290 PaperMemory local artifact. Hybrid retrieval metrics. `reports/final/results/hybrid_`
291 `metrics.md`.

292 PaperMemory local artifact. Node 6 verified chat and evidence UI. `reports/final/results/`
293 `chat_ui_packet.md`.

294 PaperMemory local artifact. Node 7 agent trace examples. `reports/final/results/agent_`
295 `trace_examples.md`.

296 PaperMemory local artifact. Node 8 robustness matrix. `reports/final/results/robustness_`
297 `matrix.md`.

298 PaperMemory local artifact. Node 8 failure gallery. `reports/final/results/failure_`
299 `gallery.md`.

300 PaperMemory local artifact. Node 9 cost-benefit stress test. `reports/final/results/cost_`
301 `benefit.md` and `reports/final/results/cost_benefit.csv`.

302 PyMuPDF. PyMuPDF project page. <https://pypi.org/project/PyMuPDF/>, source checked
303 for Node 10.

304 Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond.
305 *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009.

306 Shi Yu, Chaoyue Tang, Bokai Xu, Junbo Cui, Junhao Ran, Yukun Yan, Zhenghao Liu, Shuo Wang,
307 Xu Han, Zhiyuan Liu, and Maosong Sun. VisRAG: Vision-based retrieval-augmented generation
308 on multi-modality documents. arXiv:2410.10594, 2024.

309 A Reproducibility Commands and Environment

310 The active repository root is `D:/codex/papermemory`. The older
311 `D:/codex/llm_paper_assis/papermemory` path is now a junction to this working tree.
312 The Node 0 environment artifact records branch `codex/papermemory-evidence-agent`, base-
313 line commit `6884d738df9d685bad4f5a673a7d13a7bc2fb691`, Python 3.12.7, Node.js v22.18.0,
314 and npm 10.9.3 (PaperMemory, 2026d).

315 The final report package was compiled with the local LaTeX helper:

```
316 python "C:\Users\Miles CUI\.codex\plugins\cache\openai-bundled\latex\  

317 0.2.2\scripts\compile_latex.py" D:\codex\papermemory\  

318 reports\final\main.tex --json
```

319 Earlier nodes recorded the following verification snapshots: Node 1 synthetic retrieval generation
320 and JSON validation; Node 4 BM25 fixture generation; Node 5 hybrid fixture generation; Node
321 6 chat and evidence UI tests; Node 7 bounded orchestrator tests; Node 8 robustness tests with 11
322 focused passes; and Node 9 cost-model tests plus report regeneration. The canonical source artifacts
323 for this report are stored under `reports/final/results/` and `eval/retrieval/results/`.

324 The package uses the official `neurips_2025.sty` downloaded from the NeurIPS 2025 style link
325 on the Call for Papers page (NeurIPS, 2025a). The report uses anonymous submission style, not
326 camera-ready or preprint mode.

327 B Interview Market Validation

328 The market evidence comes from informal interviews with 10 postgraduate interviewees. The sam-
329 ple is small, so the results are treated as early intent evidence rather than paid adoption data.

330 **Questionnaire protocol.** Each interview used the same six prompts:

- 331 1. What is your current research stage, and how often do you read PDFs?
- 332 2. What is your current literature reading and citation-checking workflow?
- 333 3. On a 1–5 scale, how likely would you be to pay if PaperMemory delivered local PDF inges-
334 tion, evidence-backed answers, visible page citations, bounded uncertainty, and smoother
335 model access?
- 336 4. Would you prefer BYOK/local-model setup or PaperMemory-provided LLM access
337 through a subscription?
- 338 5. Which packaging would you prefer: free BYOK with limited groups, 10 USD/month with
339 a lower weekly managed-model limit, or 20 USD/month with a higher weekly managed-
340 model limit?
- 341 6. What risk would stop you from relying on the tool for research writing?

342 **Aggregate response.** Across the 10 postgraduate interviewees, the mean conditional purchase
343 likelihood was 4.2/5 if PaperMemory delivered the described workflow, and 70% preferred
344 PaperMemory-provided API/model access through a subscription over BYOK-only access. The
345 report frames that paid option as provider-compliant API relay with caps, not consumer-account re-
346 sale. The report does not claim participant-level raw data, paid conversion, retention, product-market
347 fit, or production margins from this interview sample.

Table 10: Interview aggregate used in the market and revenue sections.

Measure	Result	Report use
Sample	10 postgraduate re- searchers	Early postgraduate researcher signal, not a representative survey.
Conditional purchase likeli- hood	4.2 / 5 mean	Supports testing a paid evidence-workflow product if the promised functions are delivered.
Managed API/model prefer- ence	70%	Supports adding monthly subscription tiers rather than offering BYOK only.
Proposed tiers discussed in re- port	Free BYOK, 10 USD/month, 20 USD/month	Proposed packaging for stress testing, not launched billing.

348 **How the report uses the result.** The main report uses the interviews to justify a commercial
349 experiment: a free BYOK tier for adoption, plus paid provider-compliant managed-access tiers for
350 users who do not want to configure model keys. The result also explains why the cost model matters.
351 If managed access is part of the product, model usage must have weekly caps, BYOK fallback above
352 cap, rate limits, or fair-use controls.

353 **Limitations.** The interviews do not measure paid conversion, retention, referral, production rev-
354 enue, support load, or willingness to keep paying after repeated use. The report therefore treats the
355 numbers as market direction, not proof of demand.

356 C Full Metric Tables

357 The retrieval summary in the main text is derived from generated CSV/Markdown artifacts under
358 eval/retrieval/results/. The aggregate rows for the keyword overlap baseline, visual API
359 baseline, BM25 baseline, and hybrid baseline are restated in Table 11.

360 The hybrid artifact reports 37 units with visual traces, 27 with BM25 traces, 27 with both traces, 10
361 visual-only units, and 0 text-only units. This supports the page-canonical fusion claim: overlapping

Table 11: Aggregate synthetic fixture metrics with corpus and path details.

Run	Setup	Metrics	Limits
Keyword overlap baseline	Command recorded in the keyword metrics artifact. Scores title+caption pages by TOKEN_RE token overlap on the same 12 golden questions.	R@1 88.9%; R@3 100.0%; R@5 100.0%; citation-page 100.0%; refusal 33.3%.	Weak deterministic same-fixture comparator only; not LLM, BM25, VisRAG, real PDF, or real corpus evidence.
Visual API baseline	Current FastAPI retrieval route over 4 synthetic papers, 12 pages, 12 golden questions, 3 refusal cases.	R@1 77.8%; R@3 100.0%; R@5 100.0%; citation-page 100.0%; refusal 33.3%.	Not a real PDF, real local corpus, or real VisRAG evaluation.
BM25 text manifest	Direct deterministic BM25 over synthetic TextManifest pages, $k_1 = 1.5$, $b = 0.75$.	R@1 88.9%; R@3 100.0%; R@5 100.0%; citation-page 100.0%; refusal 100.0%.	Synthetic text fixture only; no route, chat, hybrid, or OCR claim.
Hybrid page fusion	Synthetic visual store plus synthetic text manifest, page-canonical RRF, $k = 60$.	R@1 77.8%; R@3 100.0%; R@5 100.0%; citation-page 100.0%; refusal 33.3%; 27 both-trace units.	Contract/provenance result, not a real-corpus improvement claim.

retriever evidence is collapsed into a shared evidence unit with rank traces instead of being exposed as duplicate citations.

D Robustness Failure Gallery and Cost Assumptions

The failure gallery records seven safe-failure snapshots: empty scope, missing scoped evidence, no-new-evidence loop, prompt-injection-like PDF text, citation drift, conflicting evidence, and low-text or scanned markers. The robustness matrix records 11 focused test passes for these fixtures (PaperMemory, 2026l,m). The expected behavior is containment and visible limits, not complete prevention of all possible prompt or citation failures.

The cost model uses the Node 9 formula:

$$\text{net savings} = (h_m - h_p - h_v)r - c_a - c_l.$$

Low, base, and high cases vary manual and verification hours. Labor conversion uses annual wage divided by 2080 hours and multiplied by 1.35. Source values for OpenAI pricing, BLS wages, Elicit, and Consensus are a 2026-06-24 planning snapshot, so they should be refreshed before any customer-facing or investor-facing use.

Table 12: Cost-model assumptions from `cost_benefit.csv`.

Scenario	Runs	Input tok.	Cached tok.	Output tok.	Labor profile
10-PDF evidence packet	3	175000	25000	17500	BLS management analysts, 65.68 USD/hr loaded.
30-PDF project evidence scan	6	690000	120000	54000	BLS data scientists, 73.08 USD/hr loaded.
Systematic-review pre-screening	10	3500000	750000	250000	BLS medical scientists, 65.29 USD/hr loaded.

PyMuPDF/MuPDF licensing remains unresolved for commercial deployment. The cost table therefore includes planning license allocations, but the paper does not convert those estimates into production margins.

The proposed 10 USD/month and 20 USD/month tiers are tier-economics assumptions rather than validated pricing telemetry. Weekly caps, BYOK fallback above cap, rate limits, and fair-use con-

380 trols are product controls for provider-compliant managed LLM API relay; the report does not as-
381 sume consumer-account pooling, unlimited inference, or production margins.

382 **NeurIPS Paper Checklist**

383 This checklist follows the NeurIPS checklist guidance (NeurIPS, 2025b). Answers are intentionally
384 conservative.

- 385 1. **Claims.** [Yes] The abstract and introduction state that performance metrics are syn-
386 thetic/local smoke evidence, and that interview evidence is early conditional intent rather
387 than paid adoption.
- 388 2. **Limitations.** [Yes] Section 8 and Appendix D cover synthetic fixtures, no OCR, no dense
389 text retrieval, no broad user study, and licensing risk.
- 390 3. **Theory.** [NA] The paper does not introduce theorems or formal proofs.
- 391 4. **Reproducibility.** [Yes] Appendix A records commit, environment, commands, and local
392 artifact paths.
- 393 5. **Datasets.** [Yes] The evaluation data are synthetic local fixtures described in the metrics
394 artifacts. Appendix B describes the small interview sample used for market proof.
- 395 6. **Compute.** [Yes] The local Python, Node, npm, and LaTeX verification commands are
396 recorded in the build notes and appendix.
- 397 7. **Ethics.** [Yes] Section 8 discusses human verification, privacy, prompt-boundary limits,
398 and licensing risk.
- 399 8. **Broader impacts.** [Yes] The system can reduce first-pass citation labor and API setup
400 friction, but misuse risk remains if users treat draft answers as verified scholarship.
- 401 9. **Safeguards.** [Yes] Deterministic guardrails are described for empty scope, missing ev-
402 idence, prompt-injection-like PDF text, citation drift, conflicting evidence, and low-text
403 markers.
- 404 10. **Licenses and assets.** [Yes] Local artifacts are listed, and PyMuPDF/MuPDF
405 AGPL/commercial-license exposure is documented as a deployment risk.